



New insights into the sea-level evolution in Corsica (NW Mediterranean) since the late Neolithic

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ABSTRACT

A new dataset of 16 ¹⁴C radiocarbon dates better constrains Relative Sea Level (RSL) changes in Corsica Island since the late Neolithic (~3.6 ka BCE). Conflicting Holocene RSL histories between the northern and southern portions of Corsica coast are presently available in literature. Here we provide new RSL data obtained by sediment coring and beachrock analysis performed along the eastern coast of the island. These data, coupled with previously available ones, were compared with the predicted RSL variations modelled by means of the sea-level equation solver SELEN. Data from different coastal sectors of Corsica indicate a coherent pattern of RSL since the late Neolithic when the RSL was placed ~3.8 m below the present mean sea level. Then sea-level rose at rate of ~2 mm a⁻¹ in the Chalcolithic period (~3.5 and ~2.3 ka BCE) followed by a significant deceleration with rates ≤0.4 mm a⁻¹ from the early Bronze Age to present time (last 4.0 ka). The total RSL variation since ~0.5 BCE does not exceed ~0.9 m. Our data are in good agreement with previous sea-level estimates made using fixed biological indicators collected in NW Corsica and derived from geo-archaeological investigations in continental France. Conversely, beachrock samples from the Bonifacio strait (southern Corsica) seem to significantly underestimate the RSL position especially between ~3.5 and ~0.5 ka BCE. Such discrepancy might reflect radiocarbon calibration issues rather than sea-level variation. These results suggest that further multiproxy investigations are fundamental to better assess the regional sea-level evolution of this archaeologically important sector of the Mediterranean.

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1. Introduction

Corsica is the third largest island in the western Mediterranean and permanent human occupation can be traced back to the 6th millennium BCE (e.g., Lewthwaite, 1990). Despite its major importance among the Mediterranean islands, the response of the ancient Corsican societies to face the Holocene coastal modifications is today almost unexplored.

This paper wants to contribute to this topic by providing new insights on timing and magnitude of Relative Sea Level (RSL) changes in Corsica since the end of the Neolithic period (~3.5 ka BCE). Corsica Island is presently separated from Sardinia (the second largest Mediterranean Island) by the narrow Bonifacio strait (~11 km in its narrowest part, maximal depth ~100 m, Fig. 1). At the Last Glacial Maximum (LGM, ~26 ka), ice sheets in the Northern Hemisphere extended above much of North America, northern Europe and Asia (e.g. Lambeck et al., 2014, Khan et al., 2015), causing a dramatic drop in sea level to 120–140 m below the present datum (e.g. Lambeck and Purcell, 2005). At the end

of the LGM (~20 ka), Corsica and Sardinia were forming a huge macro-island (measuring about 35,000 km²) placed in the middle of the western Mediterranean (Lambeck and Purcell, 2005, Fig. 1). Several studies investigated the postglacial RSL sea level changes in Sardinia (De Muro and Orrù, 1998; Orrù et al., 2004, 2011, 2014; Antonioli et al., 2007, Porqueddu et al., 2011) while Corsica was comparatively less investigated (Nesteroff, 1984; Laborel et al., 1994). Changes in RSL are the net effect of contributions from eustatic, isostatic (glacio and hydro), tectonic and local factors, all of which have different response timescales (e.g. Peltier, 2004, Khan et al., 2015). At a global scale, last deglaciation produced very rapid sea level rise until ~7.0 ka ago (e.g. Lambeck et al., 2014). This resulted in the progressive development of the Bonifacio strait and the separation of Corsica and Sardinia. Conflicting sea-level data are presently available in literature (Vacchi et al., 2016b) resulting in uncertainties regarding the timing and magnitude of this paleogeographical change.

Laborel et al. (1994) reconstructed the late Holocene RSL changes in the northern (Cap Corse) and western (Scandola) sectors of Corsica island (Fig. 1) using the *Lithophyllum byssoides* fossil rims, generally regarded to be precise RSL indicators (e.g., Rovere et al., 2015; Vacchi et al., 2016b). They provided evidence that RSL was at ~–1.5 m at

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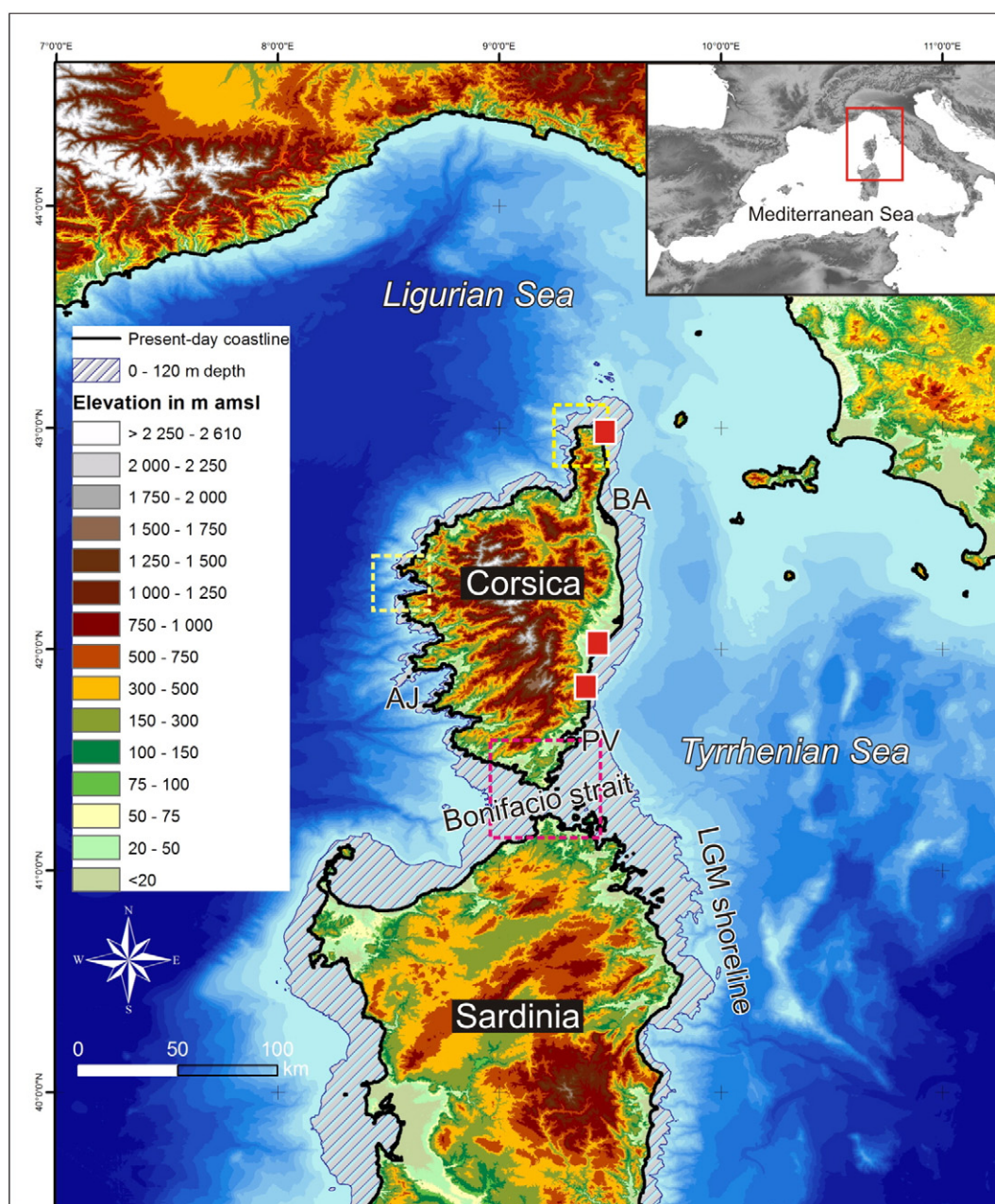


Fig. 1. Geographical position of Corsica and northern Sardinia in the western Mediterranean and approximate location of the paleoshoreline at the Last Glacial Maximum (LGM). Red squares indicate the position of the sites investigated in this study. Yellow dashed squares indicate the sites investigated by Laborel et al. (1994). Pink dashed square indicates the sites investigated by Nesteroff (1984) and De Muro and Orrù, 1998. BA is Bastia, PV is Porto Vecchio and AJ is Ajaccio.

~2.0 ka BCE rising to ~–1 m at ~1.0 ka BCE. The RSL variation since the Roman Age (i.e. 0.15 ka BCE) was, according to this study, within ~0.7 m. Nesteroff (1984); De Muro and Orrù (1998) and subsequently Lambeck et al. (2004) used a large dataset of beachrock samples to analyse the postglacial RSL changes in southern Corsica and Northern Sardinia. The large majority of these samples were collected in the wide area of the Bonifacio Strait at depth ranging between –17 and –0.3 m below the present mean sea level (msl). Beachrocks (e.g. Mauz et al., 2015) were often used for Holocene RSL reconstructions in the Mediterranean Sea (e.g., Desruelles et al., 2009; Vacchi et al., 2012b; Ozturk et al., 2015). However, their precision as RSL indicators is dependent on the analysis of cement fabric and mineralogy information as well as on sediment bedding information (Voudoukas et al., 2007; Mauz et al., 2015). Without this information, the vertical error bar

of beachrock-derived RSL indicators is large, because their formation environment ranges from slightly subtidal to supratidal (Mauz et al., 2015; Rovere et al., 2016). Beachrocks from southern Corsica yielded radiocarbon ages spanning the last 9.0 ka (Nesteroff, 1984; De Muro and Orrù, 1998). Even if affected by large error bars (Lambeck et al., 2004), they describe a different RSL history with respect to the one proposed by Laborel et al. (1994) in northwestern Corsica. Notably, in Roman Age, beachrocks indicate a RSL between ~–4 and ~–2 m. Even if 1 m of upper incertitude is added to these samples (as suggested by Lambeck et al. (2004)), however the southern Corsica RSL history remained significantly below the one from the northern coast. The study presented here contribute to this topic with new RSL data obtained by sediment coring and new beachrock analysis performed along the eastern Corsica coast.

2. Methods

2.1. Study area

With an area of $\sim 8640 \text{ km}^2$, Corsica is the third largest island of western Mediterranean (after Sicily and Sardinia). Western and the eastern coasts have different geomorphological setting. The first is generally characterized by rocky shorelines interrupted at the river mouths by coastal plains of limited extent. The topography of eastern Corsica coast is generally flat, with large coastal plain formed by the main rivers of the Island (Golo, Solenzara and Tavignano, [Forzoni et al., 2015](#)) and several large brackish lagoons and coastal saltmarshes located between Bastia and Porto Vecchio ([Fig. 1](#)). Tidal range is small and does not exceed 0.5 m ([SHOM, 2012](#)).

The western Mediterranean is a tectonically complex area where two small oceanic basins (Tyrrhenian and Liguro-Provençal back-arc basins) occur along the Nubia-Eurasia convergent margin and are separated by the Corsica-Sardinia rigid continental block ([Fig. 2](#), [Jolivet et al.,](#)

[2008](#); [Faccenna et al., 2014](#)). Sardinia-Corsica microplate played a major role in the geodynamic evolution of the present-day Western Mediterranean during Cenozoic. However, analysis of MIS 5e shorelines elevation (ranging between ~ 3 and ~ 10 m above the modern msl) indicates a general tectonic stability of Corsica and Sardinia since the last interglacial ($\sim 125 \text{ ka}$ [Ferranti et al., 2006](#); [Antonioli et al., 2009](#); [Vacchi et al., 2016b](#); [Fig. 2](#)). Further, GPS-derived vertical velocities indicate zero to weakly positive on-going vertical motion in on Corsica island ([Serpelloni et al., 2013](#), [Fig. 2](#)).

During the LGM Corsica and Sardinia formed a huge macro-island ([Fig. 1](#)). Despite the progressive post glacial sea level rise, the two islands remained strategically positioned (transition from Liguria-Tuscany/South France and North Africa) as is demonstrated by the long term human presence both in Corsica and Sardinia since the Mesolithic ($\sim 6.0 \text{ ka BCE}$, e.g., [Lewthwaite \(1990\)](#)). Because of this, the study area is very suitable to investigate RSL rise since the LGM because of the presence of a range of different markers (biological, sedimentary and archaeological).

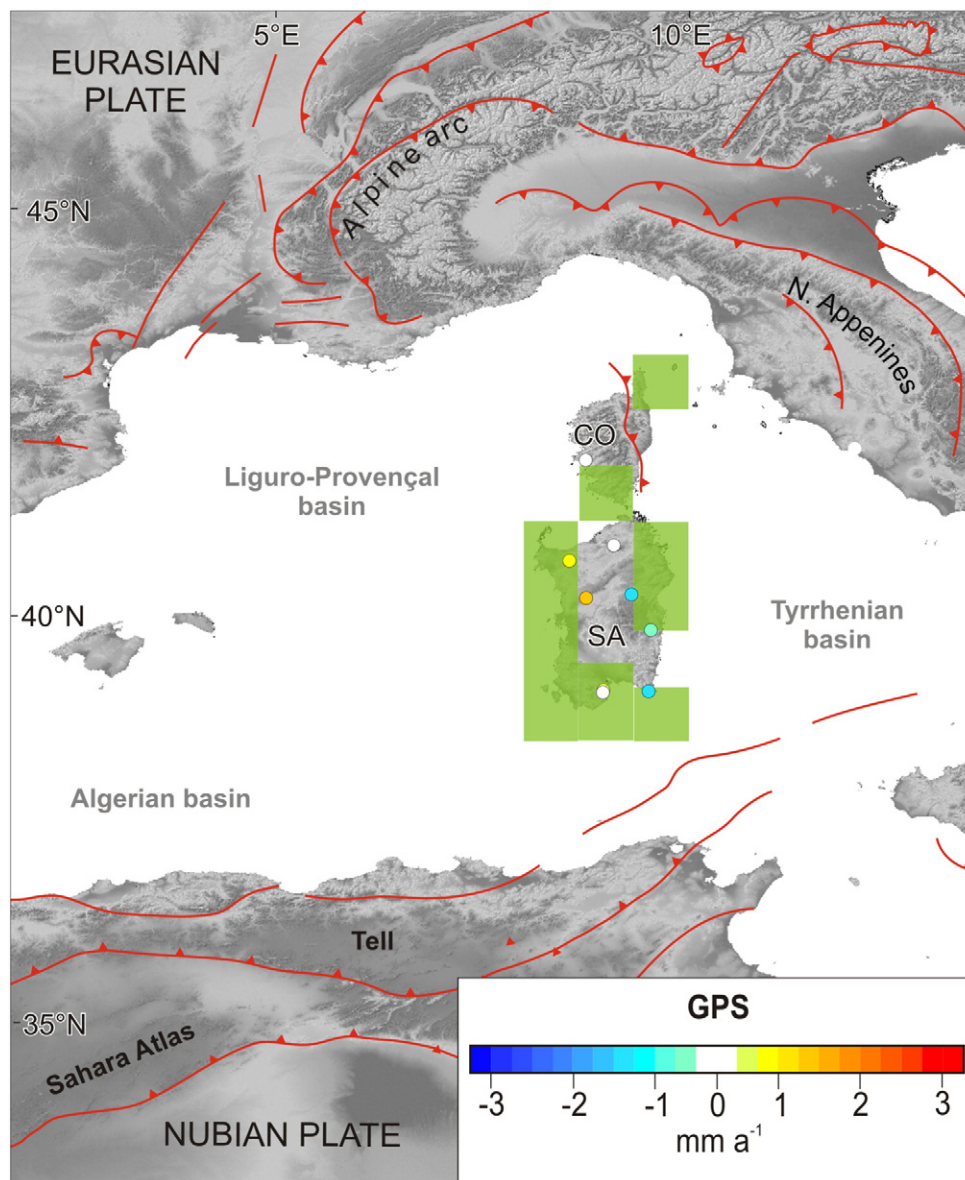


Fig. 2. Tectonic framework of the study area (modified after [Vacchi et al., 2016b](#)). Green squares indicate the sites where a long term tectonic stability (i.e. $\leq \pm 0.04 \text{ mm a}^{-1}$) is reported according to the elevation of the MIS 5e shorelines ([Ferranti et al., 2006](#), [Antonioli et al., 2009](#)). Dots indicate the on-going GPS vertical movement along the Corsica and Sardinia coasts ([Serpelloni et al., 2013](#)).

2.2. Palaeo-environmental reconstruction from sediment cores

Two 50-mm diameter vibracores were drilled in the del Sale lagoon and in Cala Francese (Fig. 1), reaching a maximum depth of -5.2 m below the mean sea level (msl). Elevation of the boreholes with respect to the msl was levelled with theodolite measurements and benchmarked to the local geodetic network, with final vertical error of ± 0.10 m. Laboratory analyses were undertaken at CEREGE in Aix-en-Provence (France).

The palaeo-environmental reconstruction was performed coupling grain size analysis with the measurement of the organic content and the determination of the mollusk assemblages found along the core.

The loss-on-ignition (LOI) measurement was performed according to Heiri et al. (2001); sediment samples of $1 \text{ g} \pm 10 \text{ mg}$ were taken at a 10-cm interval throughout the core. After drying at 105°C to constant weight, the samples were heated to 550°C for 4 h in order to estimate the organic content.

For grain-size determinations, samples were taken at 5 cm intervals. Many displayed fine particles, with general size below $200 \mu\text{m}$. Organic matter was removed for laser-diffraction particle size study. The grain-size distribution was measured using a Beckman Coulter LS 13 320 laser granulometer with a range of 0.04 to $2000 \mu\text{m}$, in 132 fractions. The calculation model (software version 5.01) uses Fraunhofer and Mie theory. For the calculation model, we used water as the medium ($RI = 1.33$ at 20°C), a refractive index in the range of that of kaolinite for the solid phase ($RI = 1.56$), and absorption coefficients of 0.15 for the 780-nm laser wavelength and 0.2 for the polarized wavelengths (Buurman et al., 1996). Samples containing fine particles were diluted, measuring between 8 and 12% of obscuration and between 45 and 70% PIDS (Polarization Intensity Differential Scattering) obscuration.

All samples were wet-sieved through a $300 \mu\text{m}$ wire mesh screen and air dried at room temperature. The portion greater than $300 \mu\text{m}$ was examined under a binocular microscope and all identifiable shells and fragments were collected and analysed in separate plastic tubes. Identification of molluscan shells was subsequently assigned to assemblages according to the Péres and Picard (1964) and Péres (1982) classification system.

2.3. Beachrock sampling and analysis

In Mignataghja (Fig. 1), two underwater transects were carried out in order to record beachrocks features (depth in relation to sea level, width and, if present, sediment bedding and structure) and to sample the beachrock slabs (Vacchi et al., 2012a; Mauz et al., 2015). Depth of the samples was recorded by averaging 2 electronic depth gauges, with an accuracy ≤ 0.5 m (Rovere et al., 2010; Vacchi et al., 2012a). Thin sections were cut in order to perform petrographic (polarizing microscope) and microstratigraphic analyses. These observations allowed

the characterization of the constituents, the presence of bioclasts as well as the type of the cements.

2.4. Chrono-stratigraphy

The age of the samples was established by a series of new 16 AMS ^{14}C radiocarbon dates taken from marine/lagoonal shell, charcoal and plant fragments as well as from remains of the Mediterranean seagrass *Posidonia oceanica* (Table 1). Because the production of atmospheric radiocarbon has varied through geological time, radiocarbon ages were calibrated to provide dates in sidereal years with a 2σ range. All samples were calibrated using CALIB 7.0. We used a laboratory multiplier of 1 with 95% confidence limits and employed the IntCal13 and Marine13 (Reimer et al., 2013) datasets for terrestrial samples and marine samples (regional ΔR is 46 ± 40 , Siani et al., 2001), respectively.

2.5. Relative sea-level reconstruction

Results of the paleo-environmental reconstructions (see Section 3.1, results) revealed facies typical of marine and lagoonal environments. Marine facies were characterized by medium to coarse sand deposited in subtidal environment; mollusk assemblages include infralittoral species (e.g. *Rissoa ventricosa*, *Bittium reticulatum*) often mixed with *Posidonia oceanica* beds, a marine seagrass living down to 40 m depth (Vacchi et al., 2016a). Due to the difficulty in establish a direct relationship with the former msl, we used samples deposited marine facies to produce marine limiting points (Vacchi et al., 2014; Engelhart et al., 2015). Reconstructed RSL must fall above these limiting points.

Coastal lagoons represent a very common feature of the eastern Corsican coast. In these lagoons, water depth is generally lower than 1 m and very seldom exceeds a few meters, consistent with the majority of northwestern Mediterranean coastal lagoons (e.g., Sabatier et al., 2010; Marco-Barba et al., 2013). In this study, we produced RSL index points (crf. Shennan, 1986; Hijma et al., 2015) using samples containing *Cerastoderma glaucum* mollusk shells, a euryhaline species living in salinities of 4–100‰ and not tolerant to significant aerial exposure (Nikula and Väinölä, 2003; Orrù et al., 2014). The associated an indicative range (crf. Hijma et al., 2015) for the *C. glaucum* samples in found lagoonal facies is from 0 to -2 msl (Vacchi et al., 2016b). However, 3 of our radiocarbon-dated samples were deposited in semi-enclosed lagoon facies. It showed sedimentological and malacological features typical of a brackish lagoonal/estuarine environment with dominance of macrofossils typical of sheltered lacustrine environments (i.e. *Cerastoderma glaucum*, *Loripes lacteus* and *Hydrobiidae* spp. Marriner et al., 2014). We measured the modern water depth of this type of brackish lagoons nearby the coring sites. It does not exceed few decimeters,

Table 1

Radiocarbon dating results. Elevation are expressed in m below the msl.

Site	Lab number	Depositional facies	Material	^{14}C age	Error	Cal CE/BCE 2σ	Elevation ($\pm$ error)	Indicative range
del Sale lagoon	Poz-71368	Semi enclosed lagoon	<i>C. glaucum</i> shell	1315	± 30	1034 CE–1249 CE	-0.56 ± 0.1 m	msl to -1 m
del Sale lagoon	Poz-65452	Semi enclosed lagoon	<i>C. glaucum</i> shell	2810	± 30	716 BCE–396 BCE	-0.9 ± 0.1 m	msl to -1 m
del Sale lagoon	Poz-65838	Semi enclosed lagoon	Charcoal	3720	± 35	2269 BCE–1985 BCE	-1.3 ± 0.1 m	msl to -1 m
del Sale lagoon	Poz-65450	Open lagoon	<i>C. glaucum</i> shell	4165	± 30	2399 BCE–2060 BCE	-1.68 ± 0.1 m	msl to -2 m
del Sale lagoon	Poz-65451	Open lagoon	<i>C. glaucum</i> shell	4175	± 30	2426 BCE–2095 BCE	-1.74 ± 0.1 m	msl to -2 m
del Sale lagoon	Poz-65840	Open lagoon	Charcoal	4065	± 35	2852 BCE–2487 BCE	-2.4 ± 0.1 m	msl to -2 m
del Sale lagoon	Poz-65453	Open lagoon	<i>C. glaucum</i> shell	4935	± 30	3373 BCE–3070 BCE	-3.24 ± 0.1 m	msl to -2 m
del Sale lagoon	Poz-74770	Marine	Shell fragments	5260	± 35	3741 BCE–3508 BCE	-4.19 ± 0.1 m	Below msl
Mignataghja	Poz-65449	Beachrock	<i>Parvicardium</i> spp. shell	5047	± 35	3541 BCE–3279 BCE	-3.8 ± 0.5 m	mhw–mlw
Cala Francese	SacA40678	Marine	<i>P. oceanica</i> leaves	2745	± 30	654 BCE–343 BCE	-1.63 ± 0.1 m	Below msl
Cala Francese	SacA40677	Marine	<i>P. oceanica</i> leaves	2620	± 30	400 BCE–160 BCE	-1.39 ± 0.1 m	Below msl
Cala Francese	SacA40672	Marine	<i>P. oceanica</i> leaves	4295	± 30	2557 BCE–2257 BCE	-3.98 ± 0.1 m	Below msl
Cala Francese	SacA40669	Marine	<i>P. oceanica</i> leaves	4245	± 30	2473 BCE–2184 BCE	-3.88 ± 0.1 m	Below msl
Cala Francese	SacA40668	Marine	<i>P. oceanica</i> leaves	3990	± 30	2141 BCE–1846 BCE	-2.8 ± 0.1 m	Below msl
Cala Francese	SacA40669	Marine	<i>P. oceanica</i> leaves	1025	± 30	1304 CE–1451 CE	-0.76 ± 0.1 m	Below msl
Cala Francese	SacA40670	Marine	<i>P. oceanica</i> leaves	3630	± 30	1653 BCE–1408 BCE	-1.84 ± 0.1 m	Below msl

being coherent with the indicative range from 0 to -1 m proposed for Mediterranean semi-enclosed lagoon facies (Vacchi et al., 2016b).

We further added to each of these index points an additional vertical error comprising: i) error associated with precision in calculating the sample altitude (± 0.10 m for our surveys), ii) core stretching/shortening error (± 0.15 m, measured in del Sale core and ± 0.10 m measured in Cala Francese core), and iii) an angle of borehole error as a function of the overburden of the sample, taken in this study as $\pm 1\%$ (Hijma et al., 2015).

In the Mediterranean microtidal setting, beachrock are considered precise RSL index point as far as they are supported by cement fabric and mineralogy analysis and, if possible, from sedimentary information (Voudoukas et al., 2007; Mauz et al., 2015). In the intertidal zone, the metastable aragonite and High Magnesium Calcite (HMC) form as irregularly distributed needles, isopachous fibres or rims and micritic cement (Desruelles et al., 2009). Samples having these characteristics have an indicative range encompassing the whole tidal range (mean high water to mean low water), thus ± 0.25 m (Solenzara Tidal Station, SHOM, 2012). The measurement error associated with the beachrock index points is not lower than 0.5 m when depths are measured with diving gauges, (Rovere et al., 2010). We further added a thickness error (Hijma et al., 2015) of 0.1 m related to the size of the beachrock sample used for the radiometric dating.

For each dated sample, RSL is estimated using the following equation:

$$RSL_i = A_i - I_i \quad (1)$$

(Shennan and Horton, 2002), where A_i is the altitude and I_i is the mid-point of the indicative range (the reference water level) of sample i , both expressed relative to the same datum, msl in our analysis.

The total vertical error is obtained by adding in quadrature individual errors according to:

$$e_i = (e_1^2 + e_2^2 + e_3^2 + e_n^2 \dots)^{1/2} \quad (2)$$

(Shennan and Horton, 2002), where e_1 , e_n represent the sources of error for each index point including the indicative range (see description above). All these errors, with the exception of the indicative range, apply also for the marine limiting points (Hijma et al., 2015).

2.6. Predictions of relative sea-level

We obtained the RSL model predictions for Corsica solving numerically the Sea-level Equation (SLE; Farrell and Clark, 1976) by means of an improved version of the open source code SELEN (Spada and Stocchi, 2007; Spada et al., 2012). SELEN assumes a laterally homogeneous, spherical, incompressible and self-gravitating Earth with Maxwell rheology. It includes the effects of rotational fluctuations on sea-level (Milne and Mitrova, 1998) and accounts for horizontal migration of shorelines following the method outlined by Peltier (2004). In all our computations, we have employed the ICE-5G model of Peltier (2004) to predict a nominal RSL curve, based on a three-layer approximation of the multi-layered viscosity profile VM2 (Table 2). To account for the uncertainties in the viscosity profiles, we have made further runs varying the viscosity profiles in each layer within a reasonable range; the

minimal and maximal viscosity values are shown in Table 2. The thickness of the elastic lithosphere has been kept fixed to 90 km in all our computations.

3. Results

3.1. Depositional environments in del Sale lagoon and Cala Francese

3.1.1. del Sale lagoon

The core in del Sale lagoon reached the depth of -5.3 m msl (-5.4 below the ground level, Fig. 3A). On the basis of the mollusk identification and the sedimentological/geochemical analyses, the following 5 sedimentary units were identified (Fig. 3A):

- Unit M is found in the lower most part of the borehole, from 4.10 to 5.25 m in depth. It consists of homogeneous medium to coarse gray sands. Modal index exhibits values ranging from 600 to 1000 μm while mean grain size is oscillating between 500 and 600 μm . The fine fraction ($<63.4 \mu\text{m}$) is poorly represented with 15 to 30% of the total of the particles. Macro and micro fauna are rare and only in the upper part of the unit M, shell fragments are identified in 4.55 m deep: they consist in large broken pieces of *Cerastoderma glaucum* and *Bittium reticulatum*. The presence of this debris indicates that deposition cannot be considered in situ and they were probably reworked by the sea. Debris of marine seagrass *Posidonia oceanica* was also noted. This unit was then deposited in marine environment. The ^{14}C dating performed on a fragment of marine shell yielded age of ~ 3.6 ka BCE (Table 1).
- Unit L is the second unit and is found between 1.80 and 4.10 m deep. It comprises very fine particles where modal index and mean grain size show similar low values, ranging from 10 to 20 μm , corresponding to clay/clayey silts. Mollusk identification reveals the significant presence of the *C. glaucum* and *Hydrobia acuta* shells. In its upper part (between -1.80 and -3 m) the unit is composed by organic clays (organic matter content is homogenous, around 5%) where large articulated *C. glaucum* shells are locally found embedded with unidentifiable plant remains and charcoals. This unit can be considered as representative of lagoonal environment, most likely connected to the sea. Four ^{14}C dates, performed on 3 articulated *C. glaucum* shells and 1 charcoal constrain the age of this unit between ~ 3.1 ka and ~ 2.1 ka BCE (Table 1).
- Unit C₁ is found between 0.65 and 1.80 m in depth. It consists in white to gray clays with variable organic matter content, decreasing from the bottom (7%) to the top (3.5%). Marks of oxidation are recorded along the sequence with orange to brown colorations. Mollusk identification indicates a significant decrease of species diversity with the exclusive presence of the *C. glaucum* lagoonal shell. Age of the Unit CL is based on three ^{14}C dates, performed on articulated *C. glaucum* shells and charcoal, well constrain the age of the unit between ~ 2.0 ka to ~ 1.2 ka BCE (Table 2). Unit CL is characteristic of a lagoonal environment where fluvial influence is gradually increasing, probably related to the deltaic progradation of the Tavignano River mouth to the South East. The low sediment accumulation (1.15 m in ca. 3.5 ka) could be indicative of a semi enclosed lagoonal environment (edification of coastal spits due to the general redistribution of the alluvial sediments coming from the Tavignano River during the historical period).
- Unit D is a thin layer (from 0.35 to 0.65 m in depth) consisting of homogeneous yellow medium to coarse sands (modal index is about 600 μm while mean grain size is comprised between 400 and 500 μm), where small round iron inclusions could indicate an anthropogenic origin of deposition. Due to the stratigraphic position, we could infer that the layer is within the past few centuries, with coincides with an era when projects were undertaken to reclaim and to drain the del Sale coastal swamps. Such works were most likely conducted during the Genovese period (1.4 to 1.8 ka CE).

Table 2

Viscosity parameters used to predict the nominal, maximal and minimal RSL curve.

ICE 5G VM2	Viscosity parameters (Pa s)		
	Upper mantle	Transition zone	Lower mantle
Nominal	$0.5 \cdot 10^{21}$	$0.5 \cdot 10^{21}$	$2.7 \cdot 10^{21}$
Maximal	$0.8 \cdot 10^{21}$	$0.8 \cdot 10^{21}$	$5.0 \cdot 10^{21}$
Minimal	$0.2 \cdot 10^{21}$	$0.2 \cdot 10^{21}$	$1.0 \cdot 10^{21}$

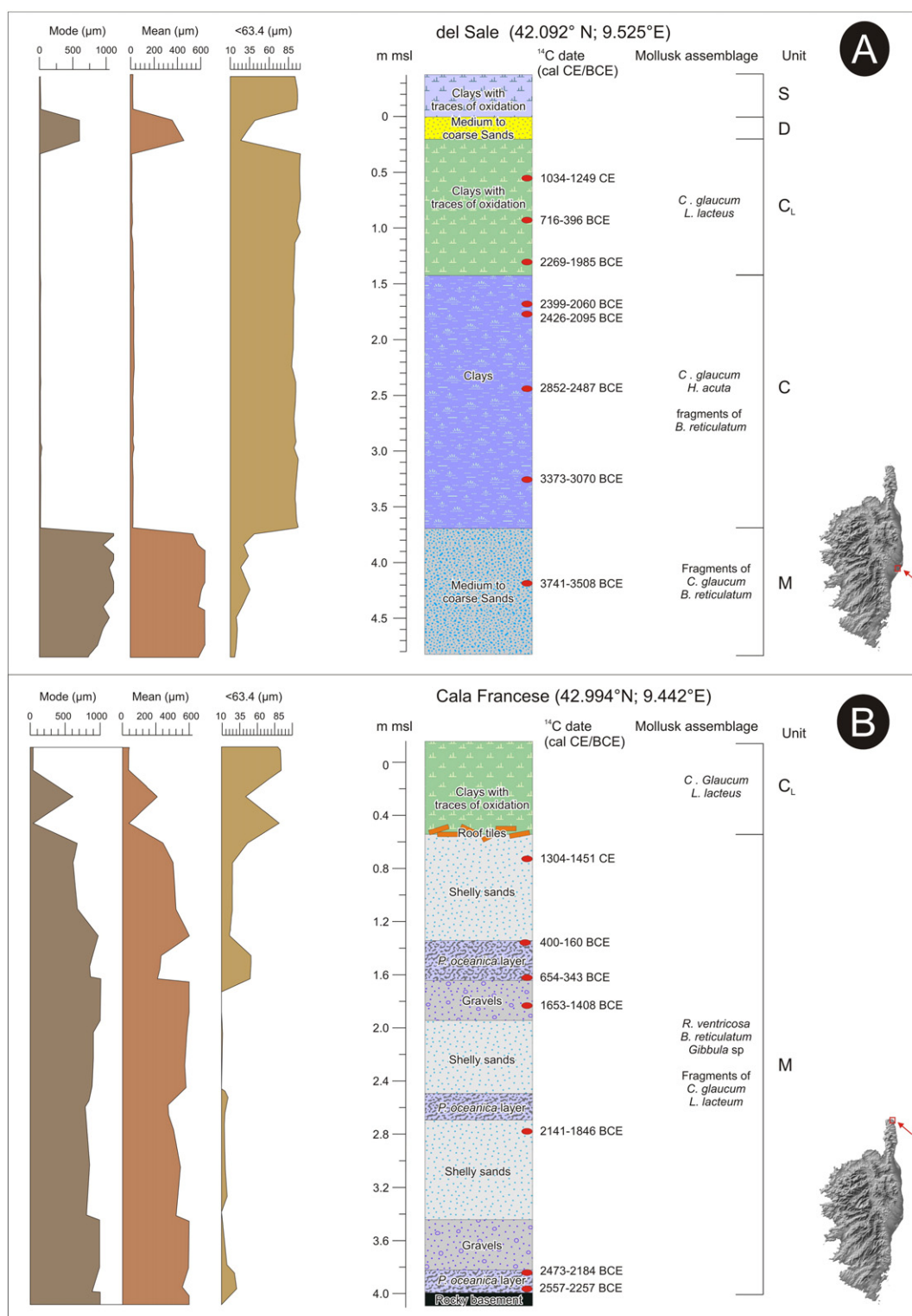


Fig. 3. A) Geographic position, stratigraphy and ^{14}C dating of the units defined for the del Sale lagoon core. B) Geographic position, stratigraphy and ^{14}C dating of the units defined for the Cala Francese core.

- Unit S is found between the surface and 0.35 m deep. It consists of white to light gray clays with numerous traces of oxidation. No identifiable mollusk fauna has been found. The characteristics of this Unit reflect the modern surface conditions and most likely correspond to the post Genovese or Second World War period.

3.1.2. Cala Francese

The core collected in Cala Francese reached the depth of -4.2 m msl (-4.32 below the ground level) reaching the basement bedrock composed by metamorphic schists (Fig. 3B). On the basis of the mollusk identification and the sedimentological/geochemical analyses, the two sedimentary units M and C_L were identified in Cala Francese.

- Unit M is found above the bedrock, from -4.0 to -0.57 m msl. Sediments mainly consist of shelly sands and gravels with intercalated beds of *P. oceanica* seagrass. The mollusk assemblage is typical of shallow marine open bay: *Bittium reticulatum* and *Rissoa ventricosa* are predominant; gastropods living on rocky shores (*Gibbula* sp.) were also found. Further, numerous fragments and eroded lagoonal shells (*Loripes lacteus*, *Cerastoderma glaucum*) are found, they clearly indicate a reworked origin of deposition. The grain size analysis reveals that two gravelly layers are found: the first between 4.0 (transition with the bedrock) and 3.48 m below the surface and the second is encountered between -1.98 and -1.63 m msl. In the upper most part of this sedimentary unit (transition with the unit C_L), large (5 cm) angular and orange remains are found in a depth of -0.68 and -0.55 m msl, they probably belong to roof tiles of a nearby structure. Seven ^{14}C dates, performed on remains of *Posidonia oceanica*, indicate that the open marine environment lasted in Cala Francese from ~ 2.5 BCE to ~ 1.3 ka CE (Table 2).
- The Unit C_L is composed by dark gray sandy clays and was found from -0.55 m msl to the surface. Mollusk assemblage reflects a semi enclosed lagoon environment where small bodies of *Loripes*

lacteus and *Cerastoderma glaucum* were exclusively found; this indicates a little and very discontinuous connections with the sea. The base of this unit shows abundant presence of orange debris (most likely roof tiles). This sedimentary unit is younger than the marine phase (i.e. ~ 1.3 ka CE, see above) but more precise age could be provided by the accurate identification of the archaeological finds that are presently under analysis.

3.2. Beachrocks morphology and microstratigraphy

Underwater surveys carried out in Mignataghja allowed identifying three distinct levels of beachrocks occurring up to ~ 300 m from the modern shoreline (Fig. 4A). The shallowest occurs between -0.3 and -0.7 m msl. A second slab outcrops at between -1.5 and -1.7 m msl and is often covered by sand. The third outcrop lies at ~ -3.8 m. This beachrock shows a very flat morphology and its thickness does not exceed 0.4 m (Fig. 4A, B). Microscope observations indicated a general high sorting grain size (ranging from fine to medium sands)

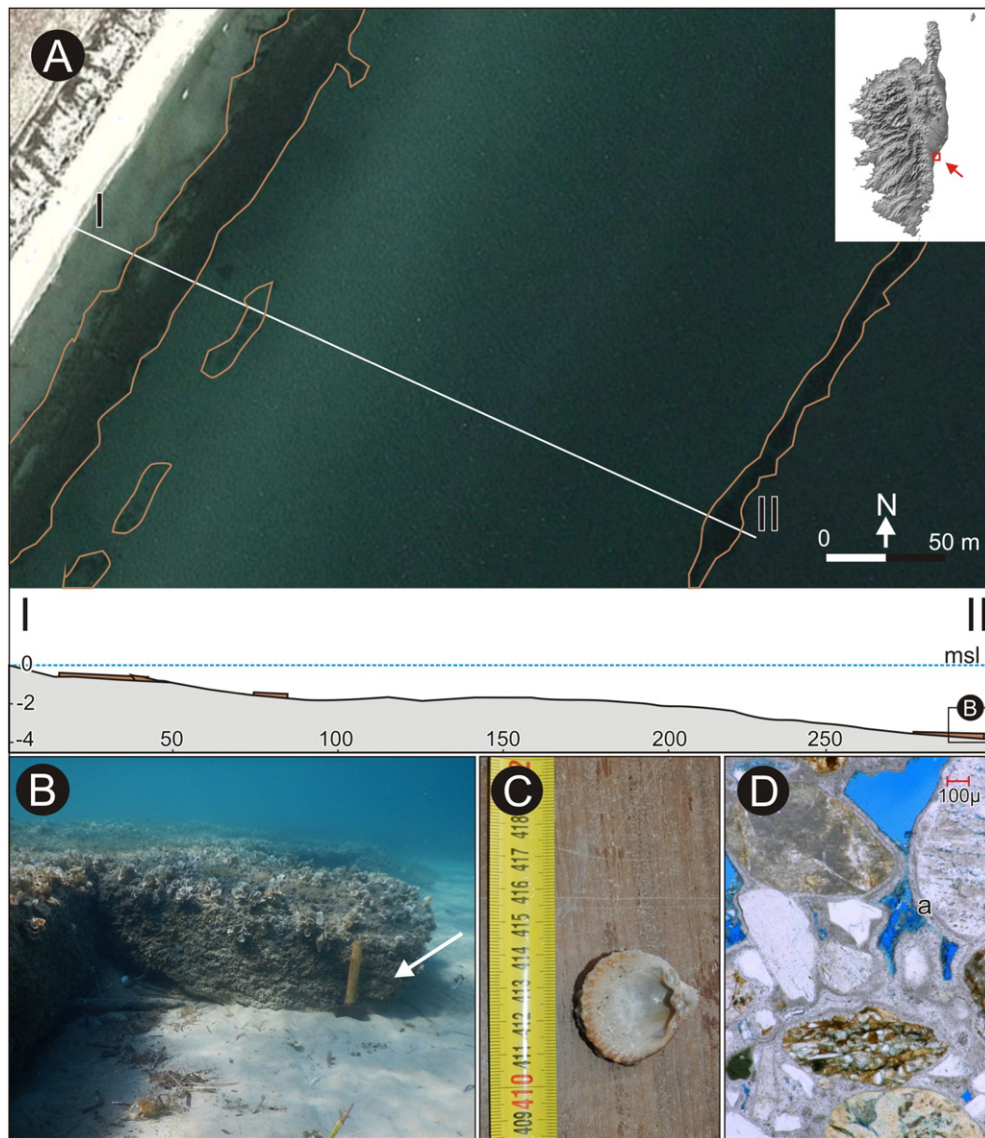


Fig. 4. A) spatial and bathymetric extent of the different beachrock levels measured in Mignataghja (source Google Earth); depth is expressed in m below the msl and distance is expressed in m from the shoreline. B) White arrow indicates the position of the sample collected from the deepest beachrock outcrop. C) *Paravicardium* spp. shell showing high degree of preservation and selected for ^{14}C dating. D) Thin section ($20\times$; plane polarized light) of the deepest beachrock sample; a) isopachous cement of radial fibrous calcite.

with presence of bioclasts, often showing a high degree of preservation (Fig. 4C). The bounding materials observed between grains are early intertidal cements and micritic filling (including internal sediments). In the thin sections, we identified an isopachous cement of radiaxial fibrous calcite (Fig. 4D), sometimes followed by geotropic deposit of internal sediment. Micritic texture, often irregularly shaped and, probably, of microbial origin, was also observed. Unfortunately, due to thickness of the cements, extraction for radiocarbon dating was extremely difficult. However, the high degree of conservation of the bioclasts in the deepest beachrock outcrop, allowed us to attempt a radiometric dating on a *Parvicardium* spp. shell (Fig. 4C). This shell, typical of lagoons and shallow infratidal zones, is generally rapidly eroded and fragmented when brought by the wave in the intertidal zone. We selected a sample that most likely did not experience lots of reworking processes having the shell a high degree of preservation (Fig. 4C). It implies a very short permanence of the dead bioclast on the beach prior to the fossilization within the beachrock formation. For this reason, we assumed that the bioclast age can reasonably corresponds to the age of the whole beachrock body. This assumption seems to be confirmed by the dating yielding a radiocarbon age of 5047 ± 35 (Table 1), coherent with the age of lagoonal index points and marine limiting derived from coring in del Sale lagoon and in Cala Francese.

3.3. RSL reconstruction

Our new radiocarbon dataset provides new insights on the RSL evolution in eastern Corsica in the last ~5.5 ka (Fig. 5). It is based on 9 index points (8 lagoonal and 1 beachrock), and 7 marine limiting points. The oldest data indicate that, in the late Neolithic (~3.6 ka BCE), RSL was above -4.2 m msl. This is in agreement with the beachrock sample

placing the RSL at -3.8 ± 0.6 m at ~3.4 ka BCE. Younger data points from the Etang del Sale document the progressive RSL rise, with rates by $\sim 2 \text{ mm a}^{-1}$, during the whole Chalcolithic period and the early Bronze Age (~3.5 to ~2.0 ka BCE). This trend is confirmed by the marine limiting points from Cala Francese (Fig. 4). At ~2.0 ka BCE, one lagoonal sample indicates the RSL was within 1 m above -1.3 m msl. Our data loosely constrain the RSL in the remaining part of the Bronze Age (Fig. 4). Conversely, one lagoonal sample indicates the RSL in the middle Iron Age (~0.5 ka BCE) was, was between -0.9 and the present msl. The youngest samples indicate that RSL variation does not exceed ~0.6 m in the last 1.0 ka. The pattern of RSL variation reconstructed with our new data seems to indicate a continuous rise of RSL up to the early Bronze Age (~2.2 ka BCE) followed by a significant decrease of the rising rates in the remaining part of the Holocene (i.e. last ~4.0 ka).

4. Discussion

We have now a robust geological record constraining the RSL variation in Corsica from the late Neolithic period (3.6 ka BCE). Our new data provide new insight about the conflicting RSL histories previously available in literature for Corsica (see Section 2). In Fig. 6 we plotted the results of this study together with the RSL reconstruction based on fossil *L. byssoides* rim (Laborel et al., 1994, re-evaluated by Vacchi et al. (2016b)) sampled along the northwest coast of Corsica and from beachrocks sampled in the Bonifacio Strait (southern Corsica, Nesteroff, 1984; De Muro and Orrù, 1998, re-evaluated by Vacchi et al. (2016b)). In this sector of the Mediterranean, no Holocene isostatic highstand is reported (Lambeck and Purcell, 2005; Stocchi and Spada, 2009; Vacchi et al., 2016b). Further, the Sardinia-Corsica block is considered tectonically stable since the last interglacial (Ferranti et al., 2006;

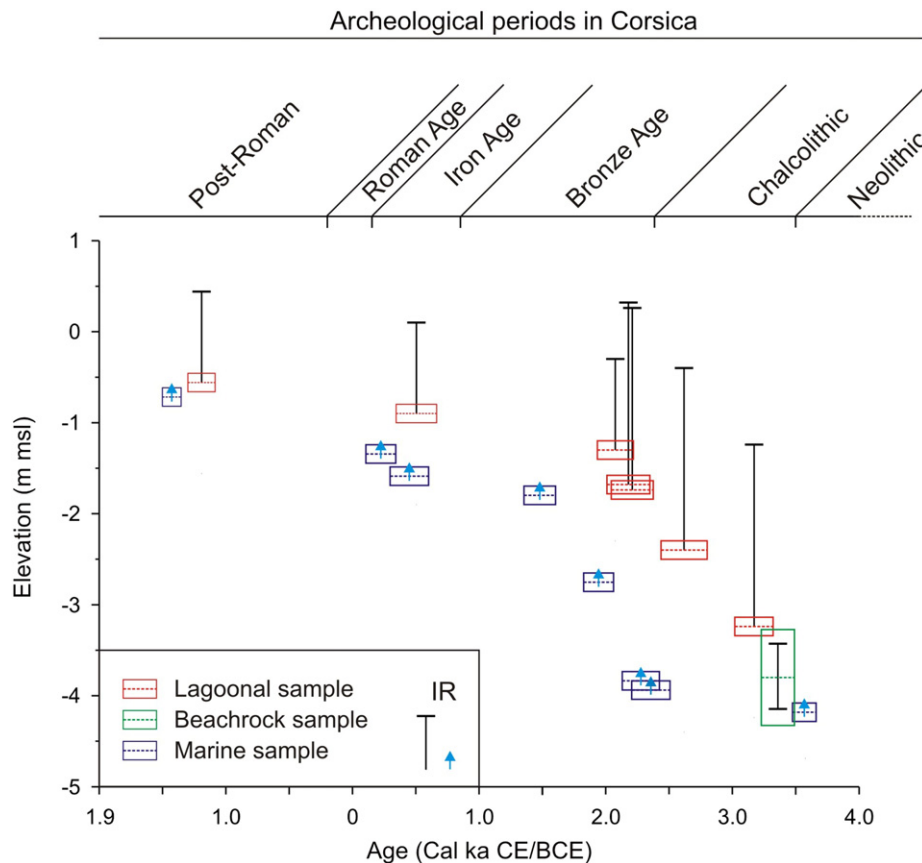


Fig. 5. Elevation ($\pm$ measurement error) and associated indicative range (IR) of the samples collected in del Sale lagoon, Cala Francese and of the beachrock of Mignataghja (see Table 1). Dashed lines indicate the sample elevation. Boxes indicate the elevation error (± 0.1 for the samples obtained from coring and ± 0.5 for the beachrock sample).

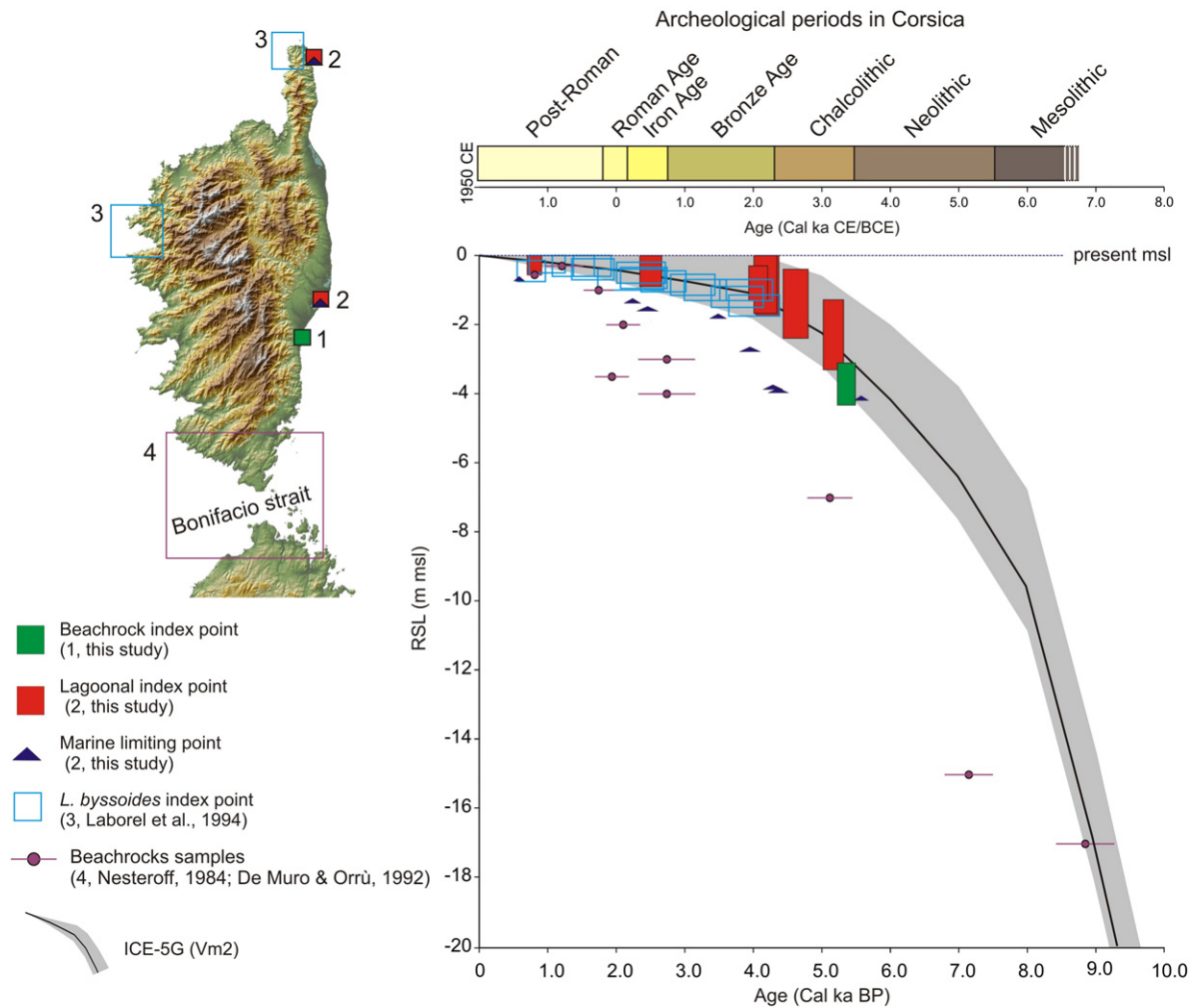


Fig. 6. Holocene RSL variations in Corsica according to different RSL proxies available for Corsica. The relative sea-level data is compared to a prediction (black line with minimum and maximum errors, see Table 2) from the SELEN model. Dimensions of boxes and lines for each point based on 2 s elevation and age errors. Y axis is forced to 0 because no RSL above the present msl is reported in the northwestern sector of the Mediterranean Sea.

Antonioli et al., 2009). For these reasons, we excluded the possibility of a Holocene RSL above the present msl by forcing the y-axis to 0 (Fig. 6).

There is a good match between the RSL reconstruction provided in this study and the one based on fossil *L. byssoides* rims, in the last 4.0 ka (Fig. 6). Further, there is a very good agreement between observed and predicted RSL changes (Fig. 6), indicating the appropriateness of ICE-5G (VM2) GIA model (Peltier, 2004) in this sector of the Mediterranean.

Conversely, there is a shift between our data and most of the beachrocks samples, especially between the Neolithic-Chalcolithic transition (~3.5 ka BCE) and the Roman Age (~0 CE, Fig. 6). Even with 1 m of upper vertical uncertainty (as suggested by Lambeck et al. (2004)), the RSL position indicated by the old beachrock samples remain significantly below the RSL reconstruction provided in this study. At ~3.3 ka BCE, the difference in elevation between the old (Nesteroff, 1984) and the new (this study) beachrock sample is 3.2 m. Between ~1.0 ka and ~0 CE, beachrocks lie up to 3 m below the sea-level position indicated by the other proxies. Only beachrocks with ages younger than 0.2 ka CE and older than 6.0 ka BCE seem to follow the RSL pattern described by both the remaining data and by the geophysical model (Fig. 5).

It is then evident that most of beachrock samples from the Bonifacio Strait significantly underestimate the RSL position, notably between the late Neolithic and the Roman Age (~3.5 ka BCE to ~0 CE). Compaction cannot account for this misfit, being the beachrocks unlikely prone to compaction (Mauz et al., 2015). Similarly, as stated above we can

exclude a tectonic influence on the current elevation of the beachrocks in our study area. A possible explanation is related to the disparities arising from radiocarbon dates of beachrock bulk cement (e.g., Voudoukas et al., 2007), especially for samples dated at the beginning of the 1980s (as most of the beachrocks of this region, Nesteroff, 1984). Multiple generations of cements have been often described in Mediterranean beachrocks (e.g. Desruelles et al., 2009; Vacchi et al., 2012a) and radiometric dating of bulk cement may result in some apparent younger age. Results of this study seem to confirm that most of beachrocks from southern Corsica have yielded a radiocarbon age younger than their actual formation near the intertidal zone. For this reason, their importance for RSL reconstruction in southern Corsica and northern Sardinia is weak.

The combined analysis of the different proxies indicate that RSL rose by ~2 mm a⁻¹ in between ~3.2 and ~2.0 ka BCE followed by a significant deceleration with rates ≤0.4 mm a⁻¹ in the last since the Bronze Age (~2.0 ka BCE). In this period, the ice equivalent meltwater input is negligible (e.g. Milne et al., 2005; Khan et al., 2015) and, in tectonically stable areas (such as Corsica, see Section 2), the total vertical land level change is related to isostatic contribution (e.g. Peltier, 2004). As consequence, our dataset better constrain the isostatic land level changes that, in Corsica, did not exceed 1.8 m for the last ~4.0 ka (Fig. 6). Our estimate of the isostatic contribution is coherent with those proposed for the continental coast of France on the basis of geo-archaeological reconstructions (Morhange et al., 2001, 2013).

ICE-5G (VM2) GIA model has proved appropriate to predict the RSL changes in this sector of the Mediterranean (e.g. Vacchi et al., 2016b). Then, we can now tentatively assess the shoreline position at the in the earliest Neolithic (~5.5 ka BCE) that, according to the model, was placed at ~−8 m msl. In Fig. 7, we attempted a reconstruction of the paleo-morphology of the early Neolithic coastline on the basis of the high-resolution modern bathymetry (Litto3D, IGN-SHOM). We are aware that caution should be use in such kind of palaeo-geographical reconstructions, especially in sites most likely affected by major progradation processes. Our reconstructions show that sectors mainly characterized by rocky coast (Cap Corse and Lavezzi Archipelago, Fig. 7A, C), did not show major coastline changes since the early Neolithic. It is clear that getting to the islands of Cavallo and Lavezzi from Mainland Corsica was significantly easier than today (Fig. 7C). However, our results do not support the hypothesis that Cavallo Island would have been accessible by walk until the Chalcolithic period (Vigne et al., 1994). Further, our reconstruction indicates a major change in the morphology of gulf of Porto Vecchio (Fig. 7B) where approximately half of the present-day gulf was land at the Neolithic. This gives new light to future archaeological investigation in this area where evidence of Neolithic frequentation was reported (e.g. Le Bourdonnec et al., 2011).

5. Conclusions

A new dataset of radiocarbon dates from coring in coastal lagoon and beachrock sampling provided new insights on the evolution of paleo sea-level in Corsica since the late Neolithic (~3.6 ka BCE). Data obtained along the coast of the island a coherent pattern of RSL histories along the Corsican coastline with negligible variability among the different sites. Former RSL reconstructions based on beachrocks sampled in the Bonifacio Strait are most likely affected by problems in the radiocarbon dating and significantly underestimate the RSL position, especially between the end of Neolithic and the Roman Age. Results of this paper call for new investigations in this sector (southern Corsica and Northern Sardinia) that should be carried out using a multiproxy approach made of sediment coring in the abundant lagoons and salt-marshes, underwater sampling of fixed biological indicators and as well as of beachrocks sampled and dated following updated protocols (e.g. Mauz et al., 2015). Such paleo-geographical reconstructions are fundamental to better investigate the numerous coastal archaeological sites described along the Bonifacio Strait and in the wider areas of southern Corsica and northern Sardinia (e.g. Biaggi et al., 1970; Wilson, 1988; Rovina, 1995). This may significantly improve the understanding of past human-environment interactions in this strategic sector of the Mediterranean.

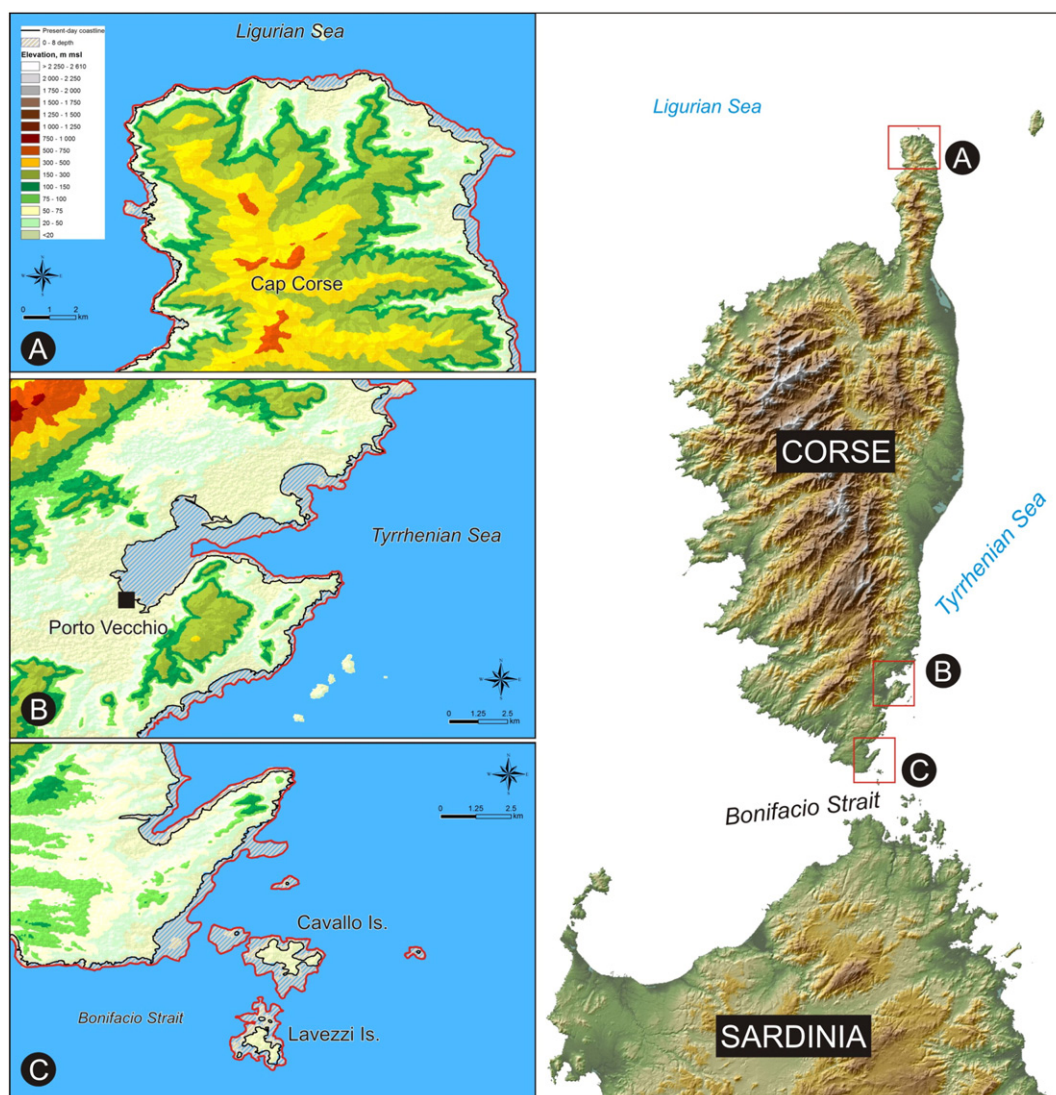


Fig. 7. Paleo-geographical reconstruction of the early Neolithic (~5.5 ka BCE) shoreline position (red line, −8 m msl) in Cap Corse (A), Gulf of Porto Vecchio (B), Lavezzi Archipelago (C).

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